UNIT 4 :

7. A system stores unique integer identification numbers using a Binary Search Tree. Write a C program to insert n values, display inorder, preorder and postorder traversals, search for a specified value, and report whether it exists. Use the output to explain why inorder traversal produces sorted values.

Source code :-

```
#include <stdio.h>

#include <stdlib.h>

struct Node

{

   int data;

   struct Node *left;

   struct Node *right;

};

struct Node* createNode(int value){

   struct Node *newNode;

   newNode = (struct Node*)malloc(sizeof(struct Node));

   newNode->data = value;

   newNode->left = NULL;

   newNode->right = NULL;

   return newNode;

}

struct Node* insert(struct Node *root, int value)

{

   if (root == NULL)

      return createNode(value);

   if (value < root->data)

      root->left = insert(root->left, value);

   else if (value > root->data)

      root->right = insert(root->right, value);

   return root;

}
```

```
void inorder(struct Node *root)
{
   if (root != NULL)
   {
      inorder(root->left);
      printf("%d ", root->data);
      inorder(root->right);
   }
}
void preorder(struct Node *root)
{
   if (root != NULL)
   {
      printf("%d ", root->data);
      preorder(root->left);
      preorder(root->right);
   }
}
void postorder(struct Node *root)
{
   if (root != NULL)
   {
      postorder(root->left);
      postorder(root->right);
      printf("%d ", root->data);
   }
}
int search(struct Node *root, int value)
{
   if (root == NULL)
```

```
        return 0;
    if (root->data == value)
        return 1;
    if (value < root->data)
        return search(root->left, value);
    else
        return search(root->right, value);
}
int main()
{
    struct Node *root = NULL;
    int n, value, key, i;
    printf("Enter number of values: ");
    scanf("%d", &n);
    printf("Enter values:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &value);
        root = insert(root, value);
    }
    printf("\nInorder Traversal: ");
    inorder(root);
    printf("\nPreorder Traversal: ");
    preorder(root);
    printf("\nPostorder Traversal: ");
    postorder(root);
    printf("\n\nEnter value to search: ");
    scanf("%d", &key);
if(search(root, key))
        printf("Value %d exists in BST\n", key);
    else
```

```
        printf("Value %d does not exist in BST\n", key);
    return 0;
}
}
```

Output :-

```
Enter number of values: 5
Enter values:
10 14 5 12 13

Inorder Traversal: 5 10 12 13 14
Preorder Traversal: 10 5 14 12 13
Postorder Traversal: 5 13 12 14 10

Enter value to search: 5
Value 5 exists in BST
```